

CMPE 372 – User Interface Design

(Section 02 – Computer Engineering)

Final Project Submission: Complete UI/UX + System-Aware Interface Design Report

The final project submission should present your work as a complete interface design and system-aware UI/UX report. This submission must combine and revise all previous assignment work into one coherent final project document.

The goal is to demonstrate not only visual interface quality, but also how the interface is logically connected to backend structures, user actions, and data flow.

Submission Deadline

June 4, 2026 23:59pm.

Late submissions will NOT be accepted.

You should both **upload your final project report to Learn** and bring **one printed copy** to the last class.

Submission Format

- Single PDF/Word Report
- Accessible Figma Link
- Source Code Files (ZIP or GitHub link) if applies
- Optional supplementary assets if needed

File Naming Convention:

GroupName_CMPE_FinalProject.pdf

Required Final Submission Sections

- Project Overview
- Problem definition
- Target users
- System purpose
- Value proposition
- Project scope
- Why the problem matters
- Limited by 1 page
- It must include:
 - project context
 - value proposition
 - system purpose

Personas

Minimum 3 personas are required. Each persona must include:

- name
- age
- role
- goals
- frustrations
- digital behavior
- typical tasks
- system needs

Personas must guide later design decisions.

User Stories

- Minimum 8 user stories
- Format: As a [user], I want to [action], so that [benefit]
- Stories should cover all major user roles

User Journey Map

- At least one complete user journey
 - Stages
 - Actions
 - Emotions
 - Pain points
 - Opportunities

Information Architecture

- It must include:
 - main modules
 - role-based navigation
 - primary navigation
 - secondary navigation
- Show how information is grouped to reduce cognitive load.

Site Map

- Provide hierarchical page structure. For example:
- Home
- → Browse
- → Detail
- → Form
- → Dashboard
- → Admin / Staff panel (if relevant)

Page Content Inventory

- For key pages define:
 - | Page | Purpose | Main Content | User Actions |
- At least 5 pages

High-Fidelity Prototype

- Minimum 6 connected Figma screens
- Must demonstrate:
 - Consistent visual hierarchy
 - Realistic UI components
 - Interactive navigation
 - Transitions and flows

Accessibility Review

- Visual accessibility
- Typography readability
- Contrast
- Keyboard accessibility
- Reduced cognitive load
- Basic WCAG alignment

Dynamic UI & Interaction Logic

- Loading states
- Dynamic content behavior
- Asynchronous interactions
- Form validation logic
- Feedback messages
- UI state changes

Database / Data Model

- Minimum 5 entities
 - entity name
 - key attributes
 - purpose
- ER Diagram
 - Primary & Foreign Keys
 - Normalization explanation
 - Relationship logic
 - Cardinality
 - Constraints

- At least 4 business rules

UI–Database Mapping

- CRUD mapping between screens and entities
- Show major Create, Read, Update, Delete operations
- Explain how screens interact with database tables
 - Format: | Screen | Entity | CRUD Action |
 - Example: | Report Form | Item | Create |

Sample SQL Logic

- Minimum 3 examples for each:
- SELECT
 - Include filtering, aggregation, grouping, and sorting examples by different type of data
- INSERT
- UPDATE
- DELETE
- JOIN
- Queries should align with your schema

Basic Front-End Implementation

- Implement at least ONE screen using HTML + CSS + JavaScript
- Basic interaction logic required
- Validation or dynamic behavior expected.

Final Reflection

- Maximum 1 page
- Biggest challenge
- What changed during the project
- What you learned about UI/UX and system thinking

Final Project Rubric (30 Points Total)

This rubric evaluates the final project as a complete UI/UX and system-aware design report, including interface design, interaction logic, database integration, usability, accessibility, and implementation thinking.

#	Evaluation Criterion	Description	Points
1	Project Definition & Scope	Problem is clearly defined, realistic, well justified, and project scope is coherent	3.0
2	Personas, User Stories & User Journey	Personas are realistic and complete; user stories reflect meaningful user goals and actions; journey is logical and coherent	4.0
3	Information Architecture & Site Map	Navigation structure, task flow, page hierarchy, and information grouping are clearly defined and understandable	3.0
4	High-Fidelity Prototype Quality	Prototype is visually polished, interaction-ready, professionally consistent, and includes coherent interaction flow	4.0
5	Accessibility Review	Accessibility principles and WCAG-related considerations are meaningfully applied and connected to design decisions	2.0
6	Dynamic UI & Interaction Logic	Dynamic behaviors, validations, loading states, asynchronous interactions, and UI feedback are clearly demonstrated	2.0
7	Database Design, ER Logic & Business Rules	Entities, relationships, keys, normalization, cardinality, and business rules are logically structured	3.0
8	UI–Database Mapping & CRUD Logic	Database operations are clearly connected with interface screens, entities, and user actions	2.0
9	SQL Logic	Queries are logically correct, diverse, and aligned with the schema; includes filtering, grouping, JOINS, and CRUD-related operations	3.0
10	Front-End Implementation	HTML/CSS/JS implementation reflects the designed interface and includes meaningful interaction logic	3.0
11	Reflection & Overall Consistency	Final reflection demonstrates learning and all project sections align coherently	1.0